Optimization Homework 4 Dai Xunlian 225040015

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1 Problem 1: Integer Linear Programming

1.1 (a) Formulation of the Problem

We are given the following integer linear programming (MILP) problem:

$$\begin{aligned}
\max \quad & 2x_1 + 3x_2 + 4x_3 + 7x_4 && (1)\\
\text{s.t.} \quad & 4x_1 + 6x_2 - 2x_3 + 8x_4 = 20 && (2)\\
& x_1 + 2x_2 - 6x_3 + 7x_4 = 10 && (3)\\
& x_1, x_2, x_3, x_4 \geq 0 && (4)\\
& x_1, x_2, x_3, x_4 \in \mathbb{Z} && (5)\\
& && (6)
\end{aligned}$$

1.2 (b) Solution Process Using Branch-and-Bound

We start by solving the linear programming relaxation of the MILP, ignoring the integer constraints. The solution to the LP relaxation is:

$$x_1 = 0, \quad x_2 = 1, \quad x_3 = 1, \quad x_4 = 2$$

The objective value is 21. However, since the solution is fractional, we perform branching on the fractional variables. The process continues until an integer solution is found.

1.3 (c) Final Solution

The final integer solution is:

$$x_1 = 0, \quad x_2 = 1, \quad x_3 = 1, \quad x_4 = 2$$

The optimal objective value is 21.

2 Problem 2: Unconstrained Optimization

2.1 (a) Stationary Points and Gradient Calculation

We are given the function:

$$f(x_1, x_2) = x_1^4 + 2(x_1 - x_2)x_1^2 + 4x_2^2$$

We compute the gradient of $f(x_1, x_2)$ by taking the partial derivatives with respect to x_1 and x_2:

$$\frac{\partial f}{\partial x_1} = 4x_1^3 + 4x_1^2(x_1 - x_2) + 2x_1^2$$

$$\frac{\partial f}{\partial x_2} = -2x_1^2 + 8x_2$$

Setting these derivatives to zero to find the stationary points, we obtain the following solutions:

$$\text{Stationary points: } (-2, 1) \quad \text{and} \quad (0, 0)$$

2.2 (b) Visualization of the Function

We visualize the function $f(x_1, x_2)$ in 3D to explore its global minimum. The surface plot shows that the function has a global minimum at the point $(-2, 1)$.

A 3D plot of the function is shown below.

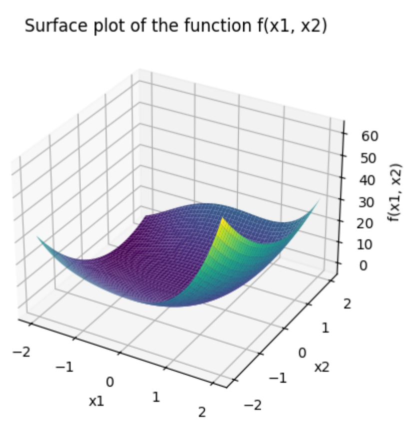

Figure 1: 3D plot of the function $f(x_1, x_2)$

3 Matrix Classification and Eigenvalues

We are tasked with classifying the following matrices and finding their eigenvalues:

3.1 Matrix A_1

The matrix A_1 is given by:

$$A_1 = \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

The eigenvalues of A_1 are:

$$\lambda_1 = 1.41421356, \quad \lambda_2 = -1.41421356, \quad \lambda_3 = -1$$

Since the matrix has both positive and negative eigenvalues, A_1 is **indefinite**.

3.2 Matrix A_2

The matrix A_2 is given by:

$$A_2 = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 1 & 2 & 5 \end{bmatrix}$$

The eigenvalues of A_2 are:

$$\lambda_1 = 5.85410197, \quad \lambda_2 = -0.85410197, \quad \lambda_3 = 4.01825745 \times 10^{-16}$$

Since A_2 has both positive and negative eigenvalues, it is **indefinite**.

3.3 Matrix A_3

The matrix A_3 is given by:

$$A_3 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 2 & 4 \end{bmatrix}$$

The eigenvalues of A_3 are:

$$\lambda_1 = 1, \quad \lambda_2 = 2, \quad \lambda_3 = 3$$

Since all the eigenvalues are positive, A_3 is **positive definite**.

3.4 Matrix A_4

The matrix A_4 is given by:

$$A_4 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ -1 & 1 & 0 \end{bmatrix}$$

The eigenvalues of A_4 are:

$$\lambda_1 = 5.67496951 \times 10^{-17} + i, \quad \lambda_2 = 5.67496951 \times 10^{-17} - i, \quad \lambda_3 = -5.35840424 \times 10^{-16}$$

Since A_4 has complex eigenvalues and a negative real eigenvalue, A_4 is **indefinite**.